

Feedback

Visualization 1:

- I'm not sure a boxplot is the right choice for this question since boxplots are more suited for showing distributions in chart. You could consider using something else
- For Chart 1, maybe get rid of the bottom half of the chart to be able to zoom into the boxplots more closely (essentially rescaling it so that it's easier to see the differences between the 3 boxplots' values)
- 0-50 just wastes space
- Change y axis starting value since right now all the boxplot all start around the 50s
- Visualization 1's bar chart uses an appropriate encoding, but the Y-axis range extends far below the actual data minimum, creating unnecessary empty space that distorts the perceived scale.
 - Remove the extra whitespace below the data in Visualization 1
- I think that having individual data points outside of the whiskers makes the plot look cluttered, so maybe just have it be the box and whisker and only plot data points outside that bound
- In their first visualization, the graph displays income groups in the order of high -> low -> medium. It might make more sense to display them in order (high -> medium -> low)
- How did you separate high/medium/low income?
- Separate by region?

Visualization 2:

- Occlusion of points. You should consider incorporating a zoom interaction into this chart (overview first, zoom and filter, details on demand). The first chart is fine, just provide interactions and maybe filter outliers from the data.
- The second visualization has a solid color scheme, except the yellow should be a bit darker or some other color, to increase contrast with the white background.
- Could be useful to do 3d plot with exam score, hours, attendance
- Reduce the number of scatterplot points through random sampling or show averages within categories

Visualization 3:

- This chart would be better represented as a scatterplot instead of a bar chart.
- Scatterplot would show lack of correlation better
- Visualization 3's Y-axis uses very small increments (0.2), which makes negligible score differences appear as dramatic performance gaps — this risks misleading the viewer about the actual effect size.
 - increase the Y-axis range in Visualization 3 to a standard scale (e.g., 0–100 or the full possible score range).
- Use a scatter plot instead of averaging for barplot

- A scatterplot is a good idea for showing the relationship between two numeric variables such as sleep or study time and exam score.
 - However, too many points at once make the chart harder to read with too many overlaps.
- Maybe include more of a “story” around the fact that sleep and scores are little correlated into a broader narrative that tells a story rather than just displaying chart

Visualization 4:

- [What are the extracurriculars? I'm not sure I understand what this chart is showing.](#)
- Visual 4 looks kinda cluttered and hard to read, maybe separate into two bar charts
 - It's really hard to read and figure out what it's trying to convey at a glance;
 - if the score is that important then it should probably be an axis instead of text or hover.
 - If you could put the hours on the bottom they were kind of hard to find or change their color
- Have the score be hover tooltip
- Add a hover feature to make the chart less cluttered
- Reduce the number of scatterplot points through random sampling or show averages within categories, and simplify Visualization 4 into a single bar chart with one combined grouping field reducing clutter.

Overall Feedbacks

- The task is clear from the presentation and I love graphics/images because they emphasize the topic.
- The visual encodings appropriate for the data types however some of the visualizations could be cleaned up to better emphasize the points.
- The use of color in the charts helped to ensure that the most important attributes were emphasized; however, the number of data points on charts 1 and 2 detracted from the visualization.
- The visualization shows all relevant data.
- The research question — whether sleep affects exam scores — is clear and well-scoped, and the bar chart and bar plot encodings are appropriate choices for the data.
- Showing every point includes the full dataset, but heavy overplotting can hide the actual trend.
- A chart that combines too many variables at once can confuse the message and make it harder to tell which factor matters most.
 - patterns across multiple variables are harder for the user to read when the plot is crowded or when one chart tries to show too many relationships at once.
- The the fact that sleep before exam and exam scores are not correlated was the major thing that caught my eye